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Classification of motor imagery by means of cortical current density estimation and Von Neumann entropy

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Abstract

The goal of the present study is to employ the source imaging methods such as cortical current density estimation for the classification of left- and right-hand motor imagery tasks, which may be used for brain–computer interface (BCI) applications. The scalp recorded EEG was first preprocessed by surface Laplacian filtering, time-frequency filtering, noise normalization and independent component analysis. Then the cortical imaging technique was used to solve the EEG inverse problem. Cortical current density distributions of left and right trials were classified from each other by exploiting the concept of Von Neumann entropy. The proposed method was tested on three human subjects (180 trials each) and a maximum accuracy of 91.5% and an average accuracy of 88% were obtained. The present results confirm the hypothesis that source analysis methods may improve accuracy for classification of motor imagery tasks. The present promising results using source analysis for classification of motor imagery enhances our ability of performing source analysis from single trial EEG data recorded on the scalp, and may have applications to improved BCI systems.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Classification of motor imagery tasks has been shown useful to form brain–computer interface (BCI) systems, to provide a new non-muscular communication path for severely disabled people. Signatures extracted from motor imagery tasks have been used to characterize subjects' intents (Wolpaw *et al* 1991, 2002, Pfurtscheller *et al* 1997, Babiloni *et al* 2000, Qin *et al* 2004, Wang and He 2004, Wang *et al* 2004, Kamousi *et al* 2005, Vallabhaneni *et al* 2005).

Recently, exploiting source analysis methods such as equivalent dipole model (Qin *et al* 2004, Kamousi *et al* 2005) and cortical imaging technique (Qin *et al* 2004) have been suggested for classifying motor imagery tasks with potential applications to BCI systems. The advantage of employing such inverse solutions versus scalp EEG recordings is that

these reconstructed source distributions may be regarded as an alternative representation of intracranial recordings that compensates the distortion and smearing effect caused by skull low conductivity and volume conduction effect (He 2004, 2005). In this work, we have used a similar idea of inverse analysis but further improved the classification results using a more advanced method integrating the distributed source imaging and statistical classification technique.

The biological basis of this work is that during unilateral hand movement imagery, an event-related desynchronization (ERD) appears on the contralateral μ rhythm. μ rhythm is an 8–12 Hz rhythm recorded over the sensorimotor cortex. Studies show that this rhythm decreases or desynchronizes both with actual or imagery movement (Pfurtscheller 1999). Therefore, left- and right-hand movement imaginations can be distinguished from each other by extracting the difference

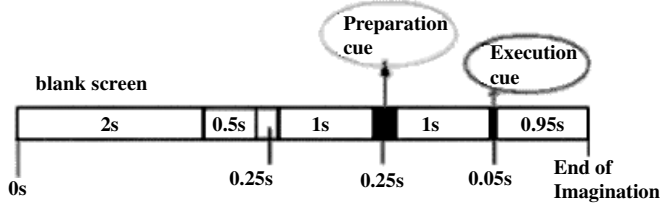


Figure 1. Time sequence of one trial epoch of the experiment.

of their corresponding μ rhythms. This forms the basis for a binary type of the BCI system.

In the present study, we use cortical current density imaging as a source analysis technique and exploit the concept of the Von Neumann relative entropy for classification. Our proposed method has been tested in a group of three human subjects. The classification results of left- and right-hand movement imaginations reconfirm the merits of source reconstruction methods for classifying motor imagery tasks which have potential applications to BCI.

2. Signal processing

2.1. Data description

The motor imagery (MI) dataset was made available by Dr Allen Osman of the University of Pennsylvania (Osman and Robert 2001, Sajda *et al* 2003). EEG data were recorded from 59 channels (placed according to the international 10/20 system) with a 100 Hz sampling rate while the subjects were seated in front of a display screen and asked to imagine either left- or right-hand movement (each subject 180 trials, 90 left, 90 right). Each trial epoch lasted for 6 s and started with a blank screen displayed for 2 s as shown in figure 1. Two noticeable timing cues are *preparation cue* and *execution cue*. During the former, which starts at 3.75 s and lasts for 0.25 s, a letter ‘L’ or ‘R’ appears on the screen indicating which hand movement should be imagined. The latter begins at 5.0 s and displays an ‘X’ for 0.05 s to tell the user to start imagination. Subjects did not have any feedback during the execution of mental tasks.

For each trial, EEG data were recorded from all 59 electrodes but since we were only interested in the activity of sensorimotor cortex, the signals from 15 channels over the sensorimotor area were used in the present study (more precisely, 15 channels after Laplacian filtering, therefore it also includes the neighbors of the electrodes of interest among these 15 channels. For more details about Laplacian filtering refer to section 2.2.1). Figure 2 shows the locations of the electrodes used in the present source analysis study on the scalp (figure 2(a)) and their schematic relative positions with the brain model (figure 2(b)). Note that the electrodes are over the scalp and figure 2(b) is just a schematic illustration.

2.2. Data preprocessing

The preprocessing step consisted of Laplacian spatial filtering, time-frequency analysis, noise normalization and independent

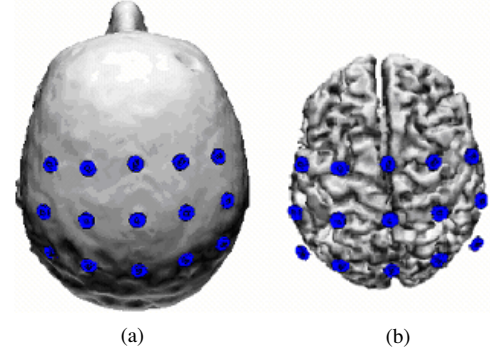


Figure 2. Layout of the channels used in this study (channels FC3, FC1, FCz, FC2, FC4, C3, C1, Cz, C2, C4, CP3, CP1, CPz, CP2, and CP4). (a) electrode positions on the scalp. (b) Their relative position with cortex.

component analysis. Details are briefly described in the following.

2.2.1. Surface Laplacian filtering. Scalp recorded EEG represents the noisy spatial overlapping of activities arising from very diverse brain regions. Spatial filter techniques attempt to accentuate localized activity and reduce diffusion in multi-channel EEG. The surface Laplacian method (Hjorth 1975, He and Cohen 1992, Babiloni *et al* 1996, He *et al* 2001, He 2004), which derives the second spatial derivative of the instantaneous spatial potential distribution, could serve for such a high-pass spatial filtering purpose. Assuming that the distances from a given electrode to its four directional neighboring electrodes are approximately equal, the surface Laplacian can be approximated by subtracting the average value of the neighboring channels from the channel of interest (Hjorth 1975).

$$M_j^{\text{Lap}} = M_j - \frac{1}{4} \sum_{k \in S_j} M_k, \quad (1)$$

where M_j is the scalp potential EEG of the j th channel, and S_j is an index set of the four neighboring channels.

2.2.2. Time-frequency filtering. Each trial lasts for 6 s, but not all time points of this 6 s period carry information about the difference between left- and right-hand MI and it is not efficient to use the whole time range for the source analysis. Subjects are told to start imagination after execution cue but they may start imagination right after preparation cue; so we have chosen the time window from 4 to 5.5 s.

In addition, the desynchronization phenomenon during motor imagery tasks is highly frequency related and mostly happens in μ rhythm. Therefore we only keep the frequency band from 8–12 Hz for source analysis because the significant difference between right- and left-hand MI appears during this frequency band. A fifth-order Butterworth filter was used for band-pass filtering. Figure 3 shows an example of the raw EEG recorded from 15 electrodes over the scalp and their corresponding signals after spatial Laplacian and temporal band-pass filtering. All signals are displayed in the chosen time interval from 4 to 5.5 s (total length of 1.5 s).

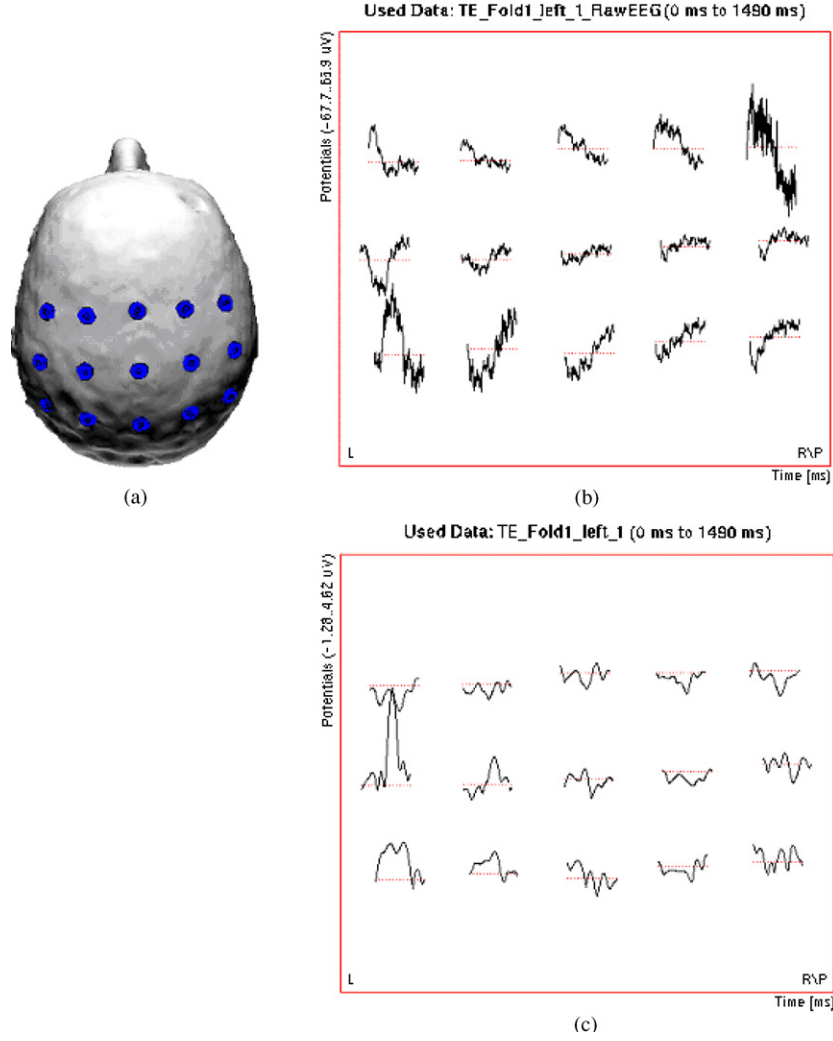


Figure 3. Scalp recordings before and after surface Laplacian filtering and band-pass filtering of one left trial. (a) The electrode positions on the scalp. (b) Time signals correspondent to each electrode. (c) Time signals after Laplacian and band-pass filtering.

2.2.3. Noise normalization. Noise normalization is one of the central tasks in preprocessing. In the present study, the filtered EEG signals from all sensors were normalized by their corresponding noise level which was estimated from certain time points taken from the histographic analysis of the data (for details, see Fuchs *et al* (1998)). The ‘20% percentile’ method was utilized to estimate the noise level. That is, within the selected time range, 20% time points with a lowest potential power are used to estimate the noise variance at each channel individually. The filtered data were then transformed to signal-to-noise-ratio (SNR) values by normalization of the signals to their corresponding noise level, yielding unit-free measurements. This was done by multiplying each EEG time series (each row), by the inverse of the square root of its corresponding diagonal element of the noise covariance matrix

$$\tilde{M}_i = \frac{1}{\sqrt{c_{ii}}} M_i, \quad (2)$$

where M_i is the spatially and temporally filtered EEG (i th channel) and $\tilde{M}_i$ consists of the SNR values after noise normalization. Note that time signal of each channel has been normalized by the corresponding noise of that channel and we

have only used the diagonal elements of the noise covariance matrix and calculated the square root of them.

2.2.4. Independent component analysis (ICA). ICA is a statistical method for finding underlying components from multidimensional data that are statistically as independent from each other as possible (Comon 1994). The goal is to find a coordinate system in which the data are maximally statistically independent

$$\tilde{M}_{n \times t} = A_{n \times n} \cdot S_{n \times t}, \quad (3)$$

where $\tilde{M}$ is the n (number of channels) by t (time points) matrix of SNR values, A is the mixing matrix and S is a matrix whose components are mutually independent. The fixed point ICA algorithm (Hyvarinen and Oja 1997) was used in the present study.

In estimating independent components no sign or energy of the independent components can be estimated, because any constant multiplying of any row of S (in equation (3)) could be canceled out by dividing the corresponding column of the mixing matrix A . However if we normalize the rows of matrix

S , then we can sort the columns of A by their corresponding norms. Sorting mixing matrix A by its column norms, independent components can be obtained in a decreasing order. In the present study, the first three components of ICA were used for source reconstruction. In other words, we have assumed that the other ICA components are mainly related to noise or artifacts so we rebuilt our data matrix $\tilde{M}$ by multiplying the first three columns of A by the first three rows of S .

2.2.5. Cortical current density estimation. As mentioned before, to classify left and right MIs, we exploit the phenomenon that during unilateral hand movement imagery, an ERD appears on the contralateral sensorimotor cortex. Unfortunately, due to smearing and distortion caused by the skull low conductivity and head volume conductor effect, the signals measured on the scalp surface do not directly indicate the brain activity underneath the scalp electrodes. Besides, although different scalp measurements must have been generated by different source configurations, different source distributions can produce quite similar scalp recorded potential distributions. Thus by recording activity at a scalp electrode, we cannot conclude that its corresponding sources are located in the area underneath it. To localize the sources we have to solve the EEG inverse problem which can only be solved by accepting some *a priori* assumptions about the sources and the volume conductor. These *a priori* assumptions are crucial, since they determine whether the solution is limited to only explaining the data or the solution actually gives neurophysiologic information about the location that signals were generated in the brain. So far several source models from single equivalent current dipole to three-dimensional (3D) current density distributions have been formulated (for a review, see He and Lian (2005)). Each of them uses different anatomical, functional and mathematical constraints. The more appropriate assumptions lead us to more realistic source estimates (Michel *et al* 2004).

In a previous work, we had explored the feasibility of directly using cortical current density (CCD) imaging to classify MIs in a human subject (Qin *et al* 2004). In the present study, the CCD model was used to map the scalp-recorded potentials onto a current dipole distribution over the cortex. In theory, CCD has a linear relationship with the scalp EEG signals since the current dipole locations are fixed. However, since the inverse procedure to estimate the CCD is in the minimum norm least-squares sense (equation (4)), the output of the CCD estimate may not be in a direct linear relationship with EEG signals. The current density estimate at any given surface grid point is a time-variant linear combination of scalp potentials. But the time course of current density estimate is not necessarily a linear combination of scalp EEG time series. The forward model we used is the three-concentric sphere head model. We have $m = 15$ scalp-recorded EEG and $n = 1497$ points over the two-dimensional surface grid of the cortex. At each of these points, there is one current dipole with fixed location and fixed orientation normal to the cortical surface at that point. The goal is to find the strength of these dipoles that

explain the scalp recordings at each time point while having the minimum overall intensity or smallest L_2 -norm.

To formulize the problem, suppose L is the $m \times n$ lead field matrix (the forward model that shows how a dipole at a given position and orientation generates signals on the scalp), j is the n by 1 vector containing cortical current density strengths (current density is defined as dipole moment per volume with unit: $\mu A \text{ mm}^{-3}$) and $\tilde{M}$ is m by 1 vector of recorded data after preprocessing steps. Solving the inverse problem in the minimum norm least-squares sense means trying to minimize the L_2 -norm of the error $[Lj - \tilde{M}]$ and the solution j at the same time. In order to do this appropriately it is necessary to introduce the regularization parameter λ and the location-weighting matrix W . It could be written as

$$\min_j \{ \|Lj - \tilde{M}\|^2 + \lambda \|Wj\|^2 \} \quad (4)$$

where the minimization is performed independently at each time point. The regularization parameter λ is necessary to balance the importance of the two terms in the minimization process. If λ is very large, the second term $\|Wj\|^2$ dominates and a large reconstruction error may occur. If λ is very small, the inverse solution may not be stable. An optimal value of λ provides an equilibrium between two terms. In this work, the L-curve criterion was used for this purpose (Hansen 1992). The matrix W is a diagonal weighting matrix used to remove the depth bias. Different source locations have different gains, where the gain is the norm of the data generated by a unit dipole. Sources with low gain need larger currents to generate the same field strength as sources with high gain and are thus suppressed as the second term punishes the overall source strength. Location weighting is used to remove the resulting bias towards high gain locations, which are usually the superficial ones (He and Lian 2005).

Figure 4 shows three examples (one for each subject) of the scalp SNR data (scalp signal map after all preprocessing steps) and their estimated cortical current density distributions at the time point with a maximum SNR. Figures 4(b), (d), (f) display the CCD values larger than 70% of the maximum CCD value, during left (b), right (d) and left (f) hand movement imagination. Figure 5 displays the power of CCD time signals at selected locations on the cortex, during left-hand movement imagination. Note that the present procedure revealed an ipsilateral activity of motor imagination, which is consistent with previous reports (Qin *et al* 2004).

3. Classification

3.1. Preprocessing for classification

After solving the inverse problem and obtaining the CCD distributions, the data we wish to classify consist of $180 \times 1497 \times 150$ matrices for each subject. The first dimension of the data matrix is 1497 because after source analysis, we have 1497 current density locations on the cortex and the second dimension is 150 because there are 150 time points. Dealing with these large matrices requires a high computation load. So it is more efficient to divide the cortex into a smaller number of regions and average the current densities over each region

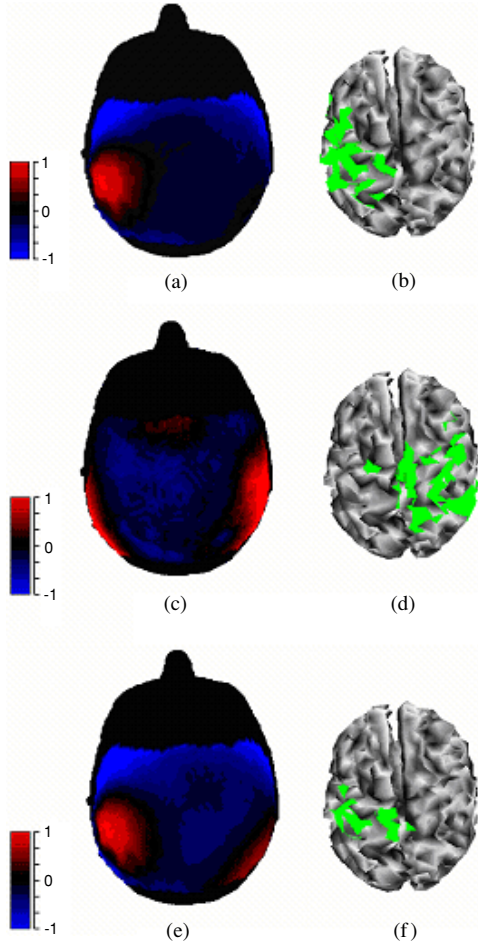


Figure 4. Examples of the scalp signal map after preprocessing ((a), (c) and (e)) and their corresponding estimated cortical current density distribution (pictures display the 30% largest current densities) ((b), (d) and (f)). The color maps on the left column code the noise-normalized scalp data.

of interest. For this purpose, we can assume a sphere that contains all of the cortical locations where CCD is estimated and cut it with equal azimuth and latitude angles from the center of the sphere. By this method, cortex could be divided to 128 regions (each azimuth angle is $2\pi/16$ rad and therefore divides the sphere into 16 wedges and each latitude angle is $\pi/8$ rad and divides each wedge into eight sectors, so the total number of regions is $16 \times 8 = 128$). Then, by summing the current densities over each region, the resulted matrix would be 128×150 which is easier to play with.

Some of these 128 regions have zero or very small current density and since we are looking for the difference between the left- and right-hand trials, these regions with low activity in all trials were discarded and only 84 regions with noticeable activity were used. Now our matrices are all 84×150 . Some of the CCD signals in this matrix are illustrated in figure 5.

As mentioned before, the desynchronization phenomenon happens during motor imagery, is highly frequency related and mostly occurs in μ rhythm (8–12 Hz). Although we have filtered the signals before CCD estimation, after all these processing and inverse procedures, it is not restricted to

8–12 Hz and a fifth-order Butterworth filter was applied again to extract 8–12 Hz signals.

3.2. Covariance matrix

Another step that can aid our classification is building the covariance matrix as defined below:

$$R = X \cdot X^T \quad (5)$$

of each trial where X is an 84×150 matrix containing time series of all 84 cortical regions. The covariance matrices represent the second-order statistical properties of each trial, which are more stable quantities for classification purposes compared to the actual time series. The diagonal elements show the signal power in each cortical location, whereas non-diagonal elements explain the cross-correlation between signals in different cortical locations. Since our data matrix X is real, all elements of R are real. Also it is easy to see that the covariance matrices are symmetric, i.e. $R_{ij} = R_{ji}$.

3.3. Von Neumann relative entropy

In order to classify the covariance matrices, we are going to use the concept of *Von Neumann relative entropy* which is a measure of distance. Entropy is a key concept of the information theory. It measures how much uncertainty there is about a random variable Y before we learn its value or in other words how much information we gain, on average, when we learn its value. The entropy of a random variable is defined to be a function of probabilities of the different possible values the random variable takes. Let Y be a discrete random variable taking a finite number of possible values $y_1, \dots, y_n$ with probabilities $p_1, \dots, p_n$. Then the Shannon entropy of Y is defined by

$$H(Y) = H(p_1, \dots, p_n) = - \sum_{i=1}^n p_i \log p_i. \quad (6)$$

Shannon relative entropy is a very useful entropy-like measure of the closeness of two probability distributions, $p(x)$ and $q(x)$, over the same index set x . The relative entropy of $p(x)$ to $q(x)$ (also known as Kullback–Leibler distance measure) is defined by Nielsen and Chuang (2000):

$$\begin{aligned} H(p(x) \| q(x)) &= \sum_x p(x) \log \frac{p(x)}{q(x)} \\ &= -H(x) - \sum_x p(x) \log q(x). \end{aligned} \quad (7)$$

In quantum mechanics, there is a definition similar to Shannon entropy but while it measures the uncertainty associated with a classical probability distribution, the *quantum* or *Von Neumann* entropy replaces the probability distributions with density operations (density matrices) and generalizes the definition of Shannon entropy to quantum states (Nielsen and Chuang 2000). For a PD matrix A , the Von Neumann entropy is defined by

$$S(A) = \text{tr}(A \log A - A), \quad (8)$$

where ‘tr’ denotes the trace of a matrix. It is also required to define the *quantum* or *Von Neumann relative entropy* which is

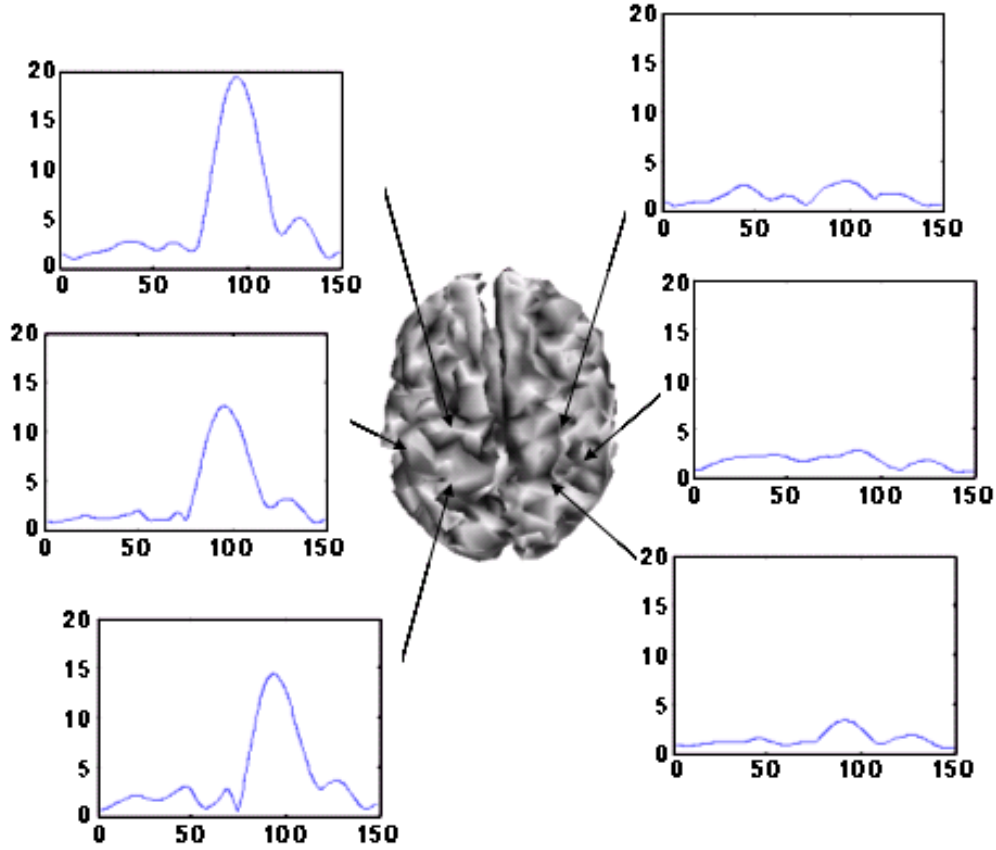


Figure 5. Examples of CCD estimate after band-pass filtering of 8–12 Hz. The horizontal axis is correspondent to time with unit ms (total 1500 ms) and vertical axis is correspondent to the CCD power with an arbitrary unit.

in a similar manner the quantum version of Shannon relative entropy. For PD matrices A and B , the relative entropy of A to B can be defined by

$$S(A\|B) = \text{tr}[(A \log A) - (A \log B) - A + B]. \quad (9)$$

Like any other function that measures the ‘distance’, the Von Neumann relative entropy has the following properties:

- (1) It is always non-negative with equality if and only if $A = B$ (for proof refer to Nielsen and Chuang (2000)):

$$S(A\|B) \geq 0. \quad (10)$$

- (2) It is jointly convex on its arguments.

Note that the relative entropy is not a symmetric function on its arguments, i.e. the distance from A to B is not equal to the distance from B to A . However, the above properties suggest that the relative entropy is a suitable measure to quantify the distance between two matrices despite its nonsymmetrical behavior. For further discussion on using relative entropy as a distance measure, we refer the readers to Georgiou (2006) and references therein.

3.4. Classification criteria

After building the covariance matrices, the dataset consists of 84×84 positive definite matrices and the Von Neumann entropy can be used to classify them. The classification criterion is as the following.

Table 1. Average relative entropy of left and right trials to left and right patterns.

	Subject 1	Subject 2	Subject 3	Average of three subjects
Mean($S(\text{left trials}\ P_L)$)	1.2773	1.0414	1.2883	1.2023
Mean($S(\text{left trials}\ P_R)$)	2.3078	1.7922	2.0931	2.0644
Mean($S(\text{right trials}\ P_L)$)	1.2978	1.1205	1.6852	1.3678
Mean($S(\text{right trials}\ P_R)$)	0.6681	0.5687	1.0454	0.7607

The dataset is divided into training and testing portions. In the training portion, two patterns, one for left and one for right trials, can be obtained by averaging their corresponding covariance matrices.

$$P_L = \text{mean}(R_{\text{Left trials}}) \quad P_R = \text{mean}(R_{\text{Right trials}}). \quad (11)$$

In testing session, for each trial, the relative entropy of the covariance matrix of the trial to the left and right patterns are obtained and compared to each other. These relative entropies can be regarded as the distance of the trial to left and right patterns. Thus, if a covariance matrix is closer to the left pattern we can conclude that it is related to a left MI and vice versa. Table 1 shows the average relative entropy of left and right trials to left and right patterns for all three subjects and figure 6 displays the classification procedure.

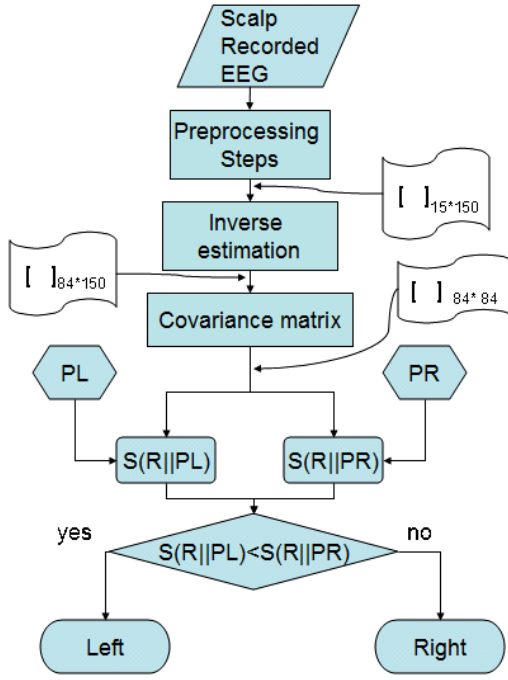


Figure 6. Classification procedure.

Table 2. Classification accuracy for three subjects.

Subject	Accuracy (%)
1	91.5
2	87.5
3	85.0
Average	88.0

4. Results

The presented inverse solution-based BCI algorithm has been tested on three human subjects. Table 2 shows the accuracy rates obtained from an 18-fold cross validation method corresponding to the three subjects and the averaged accuracy rate. In a K -fold cross validation method the dataset is divided into K equal sized folders and the test is repeated K times each time using one of the folders as a test folder and the others as a training set. Then the average classification rate across all K times is computed as the final result. This way we can make sure that the final result is not dependent on the way we choose testing and training folders and each trial gets to be in the test set exactly once. The results presented in table 2 are obtained from all 180 trials of each subject without rejecting any bad trial. The maximum accuracy rate is 91.5% for subject 2 and the average accuracy is 88.0%.

Table 3 compares the accuracy results of the same three subjects obtained by different algorithms. The first column displays the results of the two-equivalent-dipole method reported before (Kamoussi *et al* 2005); the second column shows the results of the Von Neumann-based classification on the covariance matrices of estimated CCD data. In the third column, in order to evaluate the effect of inverse estimation, the same method of the entropy-based classification has been

Table 3. Classification accuracy for three subjects obtained by different algorithms.

Subject	Two-equivalent-dipole analysis (%)	Entropy-based classification on CCD distribution (%)	Entropy-based classification on scalp recorded data (%)
1	84.4	91.5	82.2
2	77.8	87.5	54.4
3	80.5	85.0	80.0
Average	80.9	88.0	72.2

applied to scalp measurements. Processing steps are exactly the same and the only difference is that this time no inverse calculation has been exploited and the covariance matrices are based on scalp-recorded data instead of estimated cortical current density distribution.

5. Discussion

In the present study, we have developed a new classification strategy based on cortical imaging and Von Neumann entropy and tested the feasibility of the proposed method for classifying motor imagery tasks in a small group of human subjects. The present results show promise to enhance the performance of classifying motor imagery from scalp EEG by incorporating EEG inverse solutions. By performing the cortical current density imaging, we convert the scalp EEG recordings to their corresponding source signals, which were then used as inputs to the Von Neumann entropy classifier.

We have previously developed EEG inverse solutions-based classification algorithms for potential brain-computer interface applications (Qin *et al* 2004) and tested the idea in a group of four human subjects using the two moving dipole solutions (Kamoussi *et al* 2005). Similar efforts by other groups have also been reported using local field potentials (Grave de Peralta *et al* 2005).

While the equivalent dipole model provides data reduction with embedded classification outcome (location of dipoles), such a highly reduced data dimension does not seem convenient for further statistical analysis. In our previous work (Kamoussi *et al* 2005), we had modeled the sources by an over-determined model of two equivalent moving dipoles. In the dipole models the *a priori* assumption is that surface measurements can be modeled by a small number of current sources in the brain. In the present work, no *ad hoc* assumption was made with regard to the number of sources. In this distributed source model each solution point is regarded as a feasible location of a current source. The number of points is much larger than the number of measurements, which represents an underdetermined inverse problem. In the present study, we have performed the CCD source analysis on the scalp recorded EEG signals of three human subjects. The classification was again based on the phenomenon that during motor imagination, a desynchronization appears in the contralateral μ rhythm. Thus, the normalized scalp data distribution should reflect an

equivalent ipsilateral dominance. Our results of estimated CCD distribution supported this notion by showing stronger activity on the ipsilateral hemisphere of the brain. To classify the left- or right-hand MIs, we have used the anatomical power distribution of the equivalent sources and employed the relative Von Neumann entropy to pull out the power distribution differences.

In the present study, the average classification rate of 88% and maximum of 91.5% were achieved in three human subjects. This result is very positive since all the 180 trials provided by the database were used without rejecting any 'bad' trial. Yet, in practice, subjects cannot always concentrate well. Sometimes they get distracted or imagine nothing. The performance would be further improved if these 'bad' trials could be rejected based on online feedback.

By comparing the classification rates of the same three subjects obtained from different methods, we can better understand the advantage of the presented method. Column 3 of table 3 shows that if we had applied the same classification method on the scalp recordings (instead of CCD) the average classification rate was about 16% less which suggests the importance of source analysis and indicates that by compensating the smearing effect of skull, the performance of motor imagery classification can be substantially enhanced.

Another important factor that plays a major role in classification is the chosen time window and frequency band. For different subjects, the clearest ERD may occur at different frequency bands and at different time points (Pfurtscheller *et al* 1997, Wang and He 2004). Also the μ rhythm may not present in all subjects. In the present study, with the help of time-frequency representation, we have selected two fixed intervals for all subjects since the focus of the present study is to develop and evaluate a source analysis approach for BCI applications. While it is beyond the scope of the present study to find an adaptive way to choose the most suitable frequency band and time window for each subject (e.g., based on training data), it should be further explored for online experiments in future investigations.

Note that the objective of the present study was to develop and evaluate the EEG source analysis-based classification method, which can be used to classify motor imagery tasks for potential BCI applications. While our data suggest that the EEG inverse solution, in particular CCD, does improve the classification accuracy as compared with the direct use of the scalp EEG, such improved accuracy accompanies an increase in computational load. Per our experience, the most computationally demanding procedure involved is the inverse computation. The other parts such as preprocessing steps and entropy calculation have little computational load. With optimal coding and appropriate selection of regularization parameter, the inverse procedure may be implemented for online BCI applications. However, it is beyond the scope of the present study to address the optimal implementation of inverse procedure for online BCI applications.

In summary, we have investigated a unique method of classifying motor imagery tasks from scalp EEG by integrating cortical current density imaging and Von Neumann entropy. The results in the three human subjects are promising,

suggesting that the present method may offer an alternative means for effective classification of motor imagination for potential brain-computer interface applications.

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References

- Babiloni F, Babiloni C, Carducci F, Fattorini L, Onorati P and Urbano A 1996 Spline Laplacian estimate of EEG potentials over a realistic magnetic resonance-constructed scalp surface model *Electroencephalogr. Clin. Neurophysiol.* **98** 363–73
- Babiloni F, Cincotti F, Lazzarini L, Millan J, Mourino J, Varsta M, Heikkinen J, Bianchi L and Marciani M G 2000 Linear classification of low-resolution EEG patterns produced by imagined hand movements *IEEE Trans. Rehabil. Eng.* **8** 186–8
- Comon P 1994 Independent component analysis, a new concept? *Signal Process.* **36** 287–314
- Fuchs M, Wagner M, Wischmann H A, Köhler T, Theißen A, Drenckhahn R and Buchner H 1998 Improving source reconstructions by combining bioelectric and biomagnetic data *Electroencephalogr. Clin. Neurophysiol.* **107** 93–111
- Georgiou T T 2006 Relative entropy and the multivariable multidimensional moment problem *IEEE Trans. Info. Theory* **52** 1052–66
- Grave de Peralta R, Gonzalez Andino S, Perez L, Ferrez P W and Milln del R 2005 Non-invasive estimation of local field potentials for neuroprosthesis control *Cogn. Process.* **6** 59–64
- Hansen P C 1992 Analysis of discrete ill-posed problems by means of the L-curve *SIAM* **34** 561–80
- He B and Cohen R J 1992 Body surface Laplacian ECG mapping *IEEE Trans. Biomed. Eng.* **39** 1179–91
- He B 2004 *Modeling and Imaging of Bioelectric Activity-Principles and Applications* (Dordrecht: Kluwer/Plenum)
- He B 2005 *Neural Engineering* (Dordrecht: Kluwer/Plenum)
- He B and Lian J 2005 Electrophysiological neuroimaging *Neural Engineering* ed B He (New York: Kluwer/Plenum) pp 221–62
- He B, Lian J and Li G 2001 High-resolution EEG: a new realistic geometry spline Laplacian estimation technique *Clin. Neurophysiol.* **112** 845–52
- Hjorth B 1975 An on-line transformation of EEG scalp potentials into orthogonal source derivations *Electroencephalogr. Clin. Neurophysiol.* **39** 526–30
- Hyvarinen A and Oja E 1997 A fast fixed-point algorithm for independent component analysis *Neural Comput.* **9** 1483–92
- Kamoussi B, Liu Z and He B 2005 Classification of motor imagery tasks for brain-computer interface applications by means of two equivalent dipoles analysis *IEEE Trans. Rehabil. Eng.* **13** 166–71
- Michel C M, Murraya M, Lantza G, Gonzalez S, Spinelli L and Grave de Peralta R 2004 EEG source imaging *Clin. Neurophysiol.* **115** 2195–222
- Nielsen M A and Chuang I L 2000 *Quantum Computation and Quantum Information* (Cambridge: Cambridge University Press) pp 501–27
- Osman A and Robert A 2001 Time-course of cortical activation during overt and imagined movements *Proc. Cognitive Neuroscience Annual Meeting* (New York, Mar. 2001)
- Pfurtscheller G 1999 *Event-Related Desynchronization* (Amsterdam: Elsevier) pp 303–25

- Pfurtscheller G, Neuper C, Flotzinger D and Pregenzer M 1997 EEG-based discrimination between imagination of right and left hand movement *Electroencephalogr. Clin. Neurophysiol.* **103** 642–51
- Qin L, Ding L and He B 2004 Motor imagery classification by means of source analysis for brain-computer interface applications *J. Neural Eng.* **1** 135–41
- Sajda P, Gerson A, Mueller K R, Blankertz B and Parra L 2003 A data analysis competition to evaluate machine learning algorithms for use in brain–computer interfaces *IEEE Trans. Neural Syst. Rehabil. Eng.* **11** 184–5
- Stark H and Woods J W 2001 *Probability and Random Processes with Applications to Signal Processing* 3rd edn (The Netherlands: Pearson Education)
- Vallabhaneni A, Wang T and He B 2005 Brain computer interface *Neural Engineering* ed B He (Dordrecht: Kluwer/Plenum) pp 85–122
- Wang T, Deng J and He B 2004 Classifying EEG-based motor imagery tasks by means of time-frequency synthesized spatial patterns *Clin. Neurophysiol.* **115** 2744–53
- Wang T and He B 2004 An efficient rhythmic component expression and weighting synthesis strategy for classifying motor imagery EEG in brain computer interface *J. Neural Eng.* **1** 1–7
- Wolpaw J R, Birbaumer N, McFarland D J, Pfurtscheller G and Vaughan T M 2002 Brain–computer interfaces for communication and control *Clin. Neurophysiol.* **113** 767–91
- Wolpaw J R, Flotzinger D, Pfurtscheller G and McFarland D J 1997 Timing of EEG-based cursor control *J. Clin. Neurophysiol.* **14** 529–38
- Wolpaw J R, McFarland D J, Neat G W and Forneris C A 1991 An EEG-based brain–computer interface for cursor control *Electroencephalogr. Clin. Neurophysiol.* **78** 252–9